

● RAG Learning

4.5 Pythagoras & Trigonometry

Pearson Edexcel GCSE Mathematics 1MA1 — Geometry and measures — spec refs G20–G23 — Papers 1 (non-calculator), 2 and 3 (calculator)

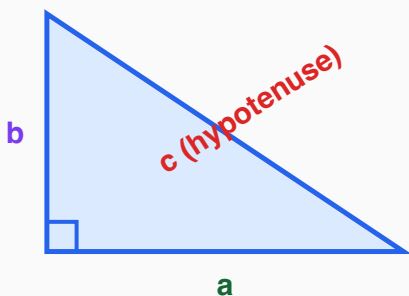
This topic is about finding missing sides and angles in triangles. On both tiers you need Pythagoras' theorem and the three trigonometric ratios in right-angled triangles in **two dimensions** (G20), and the exact values of sin, cos and tan for the special angles (G21). On the Higher tier, G20 extends to **three-dimensional** figures and to general triangles, and G22 and G23 add the sine rule, the cosine rule and $\frac{1}{2}ab \sin C$. The 1MA1 specification says you must *know* all of these formulae — none of them is given in the exam.

4.5.1 Pythagoras' Theorem (G20)

Pythagoras' theorem: in a **right-angled triangle**, the square of the **hypotenuse** (the longest side, opposite the right angle) is equal to the sum of the squares of the two shorter sides.

$$a^2 + b^2 = c^2$$

where c is the hypotenuse.



Finding the hypotenuse (square, *add*, square root):

$$c = \sqrt{a^2 + b^2}$$

Finding a shorter side (square, *subtract*, square root):

$$a = \sqrt{c^2 - b^2}$$

Worked Example: A right-angled triangle has shorter sides 7 cm and 24 cm. Work out the length of the hypotenuse.

$$c^2 = 7^2 + 24^2 = 49 + 576 = 625$$

$$c = \sqrt{625} = \mathbf{25 \text{ cm}}$$

Worked Example: A ladder 6.1 m long leans against a vertical wall. Its foot is 1.1 m from the wall on level ground. How far up the wall does the ladder reach?

The ladder is the hypotenuse, so subtract: $h^2 = 6.1^2 - 1.1^2 = 37.21 - 1.21 = 36$

$$h = \sqrt{36} = \mathbf{6 \text{ m}}$$

Pythagorean Triples

Whole-number sets that satisfy $a^2 + b^2 = c^2$ are worth recognising for Paper 1.

Triple	Some multiples
3, 4, 5	6, 8, 10 15, 20, 25 1.5, 2, 2.5
5, 12, 13	10, 24, 26 15, 36, 39
8, 15, 17	16, 30, 34
7, 24, 25	14, 48, 50

Is the Triangle Right-Angled?

By the **converse** of Pythagoras, if the longest side c satisfies $a^2 + b^2 = c^2$, the angle opposite c is a right angle. If $a^2 + b^2 \neq c^2$ the triangle is not right-angled.

Worked Example: Show that a triangle with sides 12 cm, 35 cm and 37 cm is right-angled.

$$12^2 + 35^2 = 144 + 1225 = 1369 \text{ and } 37^2 = 1369.$$

These are equal, so the triangle **is right-angled**, with the right angle opposite the 37 cm side.

Length of a Line Segment

The horizontal and vertical differences between two points form the shorter sides of a right-angled triangle:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Worked Example: Work out the distance between $P(-4, 1)$ and $Q(8, 6)$.

Horizontal difference = $8 - (-4) = 12$; vertical difference = $6 - 1 = 5$

$$d = \sqrt{12^2 + 5^2} = \sqrt{169} = \mathbf{13 \text{ units}}$$

Heights of Isosceles Triangles

The line of symmetry of an isosceles triangle splits it into two congruent right-angled triangles, so Pythagoras gives the perpendicular height.

Worked Example: An isosceles triangle has sides 17 cm, 17 cm and a base of 16 cm. Work out its area.

Half the base is 8 cm. Height = $\sqrt{17^2 - 8^2} = \sqrt{225} = 15 \text{ cm}$

$$\text{Area} = \frac{1}{2} \times 16 \times 15 = \mathbf{120 \text{ cm}^2}$$

Exam Tip: Find the hypotenuse first. Adding the squares when you need a shorter side is the most common mistake. Sense-check: the hypotenuse must be the longest side.

4.5.2 Trigonometric Ratios in Right-Angled Triangles (G20)

Label the sides relative to the angle θ you are working with:

- **Hypotenuse (H)** — the longest side, opposite the right angle
- **Opposite (O)** — the side opposite θ
- **Adjacent (A)** — the side next to θ that is not the hypotenuse

For a given angle the ratios are always the same, because all right-angled triangles with that angle are **similar**.

SOH

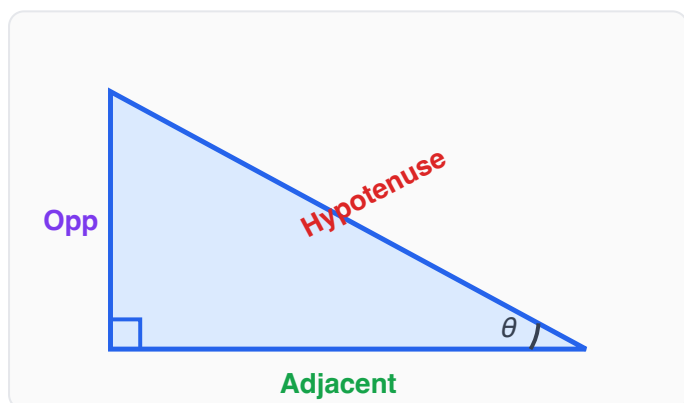
$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

CAH

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

TOA

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$



Method

1. Label O, A and H relative to the angle.
2. Identify the two sides involved: the one you know and the one you want.
3. Choose sin, cos or tan to match those two sides.
4. Substitute and rearrange. To find an **angle**, use $\sin^{-1}$, $\cos^{-1}$ or $\tan^{-1}$.

Worked Example: Unknown on top: In triangle XYZ, angle Y = 90° , angle X = 37° and XZ = 24 cm. Work out YZ.

YZ is opposite 37° and XZ is the hypotenuse, so use sin: $\sin 37^\circ = \frac{YZ}{24}$

$$YZ = 24 \sin 37^\circ = 14.44 \dots \approx \mathbf{14.4 \text{ cm}}$$

Worked Example: Unknown underneath: The side opposite a 32° angle is 50 m and the adjacent side is x . Work out x .

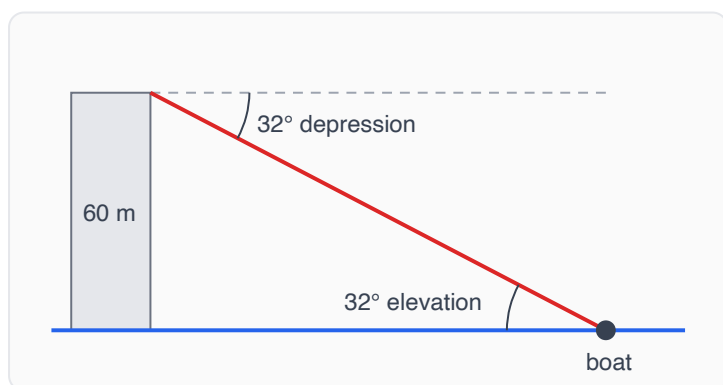
$$\tan: \tan 32^\circ = \frac{50}{x}, \text{ so } x = \frac{50}{\tan 32^\circ} = 80.01 \dots \approx \mathbf{80.0 \text{ m}}$$

Worked Example: Missing angle: The side opposite θ is 9 cm and the hypotenuse is 14 cm. Work out θ .

$$\sin: \sin \theta = \frac{9}{14}, \text{ so } \theta = \sin^{-1}\left(\frac{9}{14}\right) = 40.005 \dots \approx \mathbf{40.0^\circ}$$

Angles of Elevation and Depression

Both angles are measured **from the horizontal** — elevation looking up, depression looking down. Because the horizontals are parallel, the angle of depression from A to B equals the angle of elevation from B to A (alternate angles).



Worked Example: From the top of a cliff 60 m high, the angle of depression of a boat is 32° . How far is the boat from the foot of the cliff?

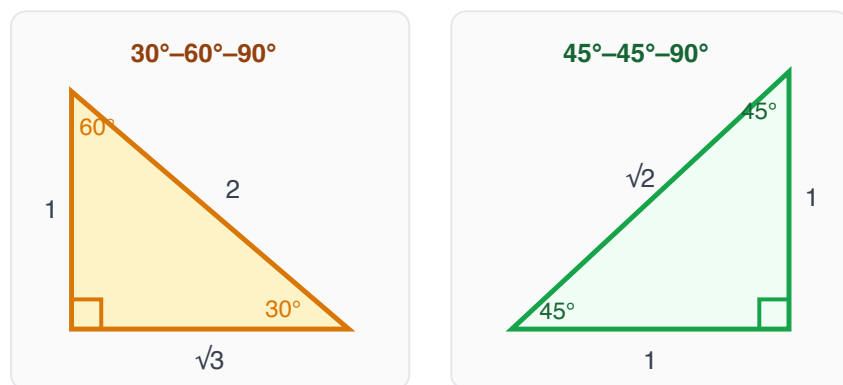
The angle of elevation of the cliff top from the boat is also 32° . $\tan 32^\circ = \frac{60}{d}$

$$d = \frac{60}{\tan 32^\circ} = 96.02 \dots \approx \mathbf{96.0 \text{ m}}$$

Exam Tip: Check your calculator is in **degrees** mode (D on the screen). Store unrounded values in the memory and round only the final answer.

4.5.3 Exact Trigonometric Values (G21)

Both tiers must know these values without a calculator. They come from two special triangles.



The 30° – 60° – 90° triangle is half of an equilateral triangle of side 2, so its height is $\sqrt{2^2 - 1^2} = \sqrt{3}$. The 45° triangle is a right-angled isosceles triangle with shorter sides 1, so its hypotenuse is $\sqrt{2}$.

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	(not required)

G21 requires sin and cos for 0° , 30° , 45° , 60° and 90° , but tan only for 0° , 30° , 45° and 60° ($\tan 90^\circ$ is undefined). $\frac{\sqrt{2}}{2}$ and $\frac{1}{\sqrt{2}}$ are equal and both are accepted.

Worked Example: Without a calculator, find the exact length of the side adjacent to 30° in a right-angled triangle with hypotenuse 10 cm.

$$\text{adjacent} = 10 \cos 30^\circ = 10 \times \frac{\sqrt{3}}{2} = 5\sqrt{3} \text{ cm}$$

Worked Example: Work out the exact value of $\cos 45^\circ \times \sin 45^\circ + \cos 60^\circ$.

$$= \frac{\sqrt{2}}{2} \times \frac{\sqrt{2}}{2} + \frac{1}{2} = \frac{2}{4} + \frac{1}{2} = 1$$

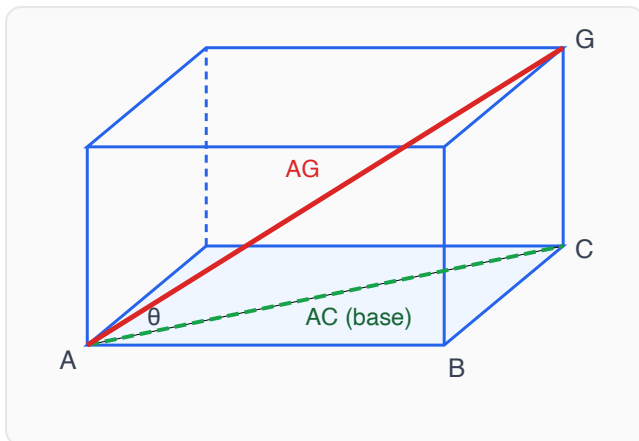
Exam Tip: Trigonometry on Paper 1 (non-calculator) will use these special angles. "Give your answer as an exact value" means leave surds such as $5\sqrt{3}$ — don't convert to a decimal.

4.5.4 Pythagoras and Trigonometry in 3D (G20) Higher

Find a right-angled triangle lying in a flat plane inside the solid, sketch it on its own, then use Pythagoras or trigonometry. Usually you need one triangle in the base and then a second, vertical triangle.

Space diagonal of a cuboid with edges a, b, c :

$$d = \sqrt{a^2 + b^2 + c^2}$$



Worked Example: Cuboid ABCDEFGH has $AB = 8$ cm, $BC = 6$ cm and $CG = 4$ cm. Work out the length of AG and the angle between AG and the base $ABCD$.

Base: $AC^2 = 8^2 + 6^2 = 100$, so $AC = 10$ cm

Triangle ACG (right angle at C): $AG = \sqrt{10^2 + 4^2} = \sqrt{116} = 10.77 \dots \approx \mathbf{10.8}$ cm

$\tan \theta = \frac{4}{10}$, so $\theta = 21.80 \dots \approx \mathbf{21.8}^\circ$

Angle Between a Line and a Plane

Drop a perpendicular from the end of the line to the plane. The angle required is between the line and its **projection** on the plane (the line from its foot to where the perpendicular lands).

Worked Example: A pyramid has a square base of side 10 cm. Its apex is vertically above the centre of the base and each sloping edge is 12 cm. Work out the height of the pyramid and the angle between a sloping edge and the base.

Half the diagonal of the base $= \frac{1}{2}\sqrt{10^2 + 10^2} = 5\sqrt{2} = 7.07 \dots$ cm

Height $= \sqrt{12^2 - (5\sqrt{2})^2} = \sqrt{144 - 50} = \sqrt{94} \approx \mathbf{9.70}$ cm

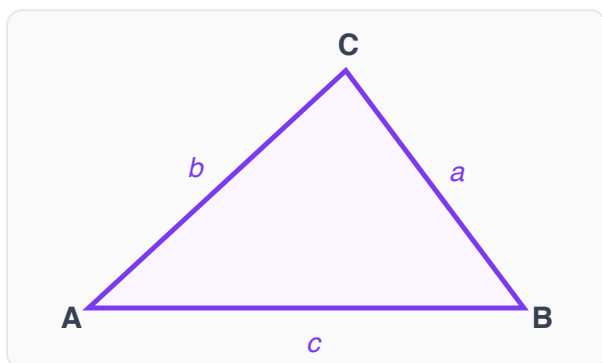
$\cos \theta = \frac{5\sqrt{2}}{12}$, so $\theta \approx \mathbf{53.9}^\circ$

Exam Tip: Draw each 2D triangle separately and mark its right angle. For a pyramid with its

apex over the centre, use *half* the base diagonal.

4.5.5 The Sine Rule (G22) Higher

For a triangle **without** a right angle, label the vertices A, B, C and the sides opposite them a, b, c .



Sine rule:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \quad \text{or} \quad \frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Use it when you know a side **and the angle opposite it**, plus one more side or angle.

Worked Example: In triangle ABC, $A = 42^\circ$, $B = 81^\circ$ and $a = 10$ cm. Work out b .

$$\frac{b}{\sin 81^\circ} = \frac{10}{\sin 42^\circ} \Rightarrow b = \frac{10 \sin 81^\circ}{\sin 42^\circ} = 14.76 \dots \approx \mathbf{14.8 \text{ cm}}$$

Worked Example: In triangle ABC, $a = 9$ cm, $b = 14$ cm and $B = 72^\circ$. Work out angle A .

$$\frac{\sin A}{9} = \frac{\sin 72^\circ}{14} \Rightarrow \sin A = \frac{9 \sin 72^\circ}{14} = 0.6113 \dots$$

$$A = \sin^{-1}(0.6113 \dots) = \mathbf{37.7^\circ} \text{ (the other solution, } 142.3^\circ, \text{ is impossible since } 142.3^\circ + 72^\circ > 180^\circ)$$

The Ambiguous Case

Since $\sin \theta = \sin(180^\circ - \theta)$, an equation like $\sin B = 0.8209$ has two solutions between 0° and 180° : $B = 55.2^\circ$ or $B = 180^\circ - 55.2^\circ = 124.8^\circ$. This can happen when you are given two sides and an angle that is *not* between them. For $a = 6$, $b = 8$, $A = 38^\circ$: $\sin B = \frac{8 \sin 38^\circ}{6} = 0.8209\dots$, and $38^\circ + 124.8^\circ < 180^\circ$, so **two** different triangles fit the information.

Exam Tip: Put the unknown on top: $\frac{a}{\sin A}$ form to find a side, $\frac{\sin A}{a}$ form to find an angle. You only ever use two of the three fractions.

4.5.6 The Cosine Rule (G22) Higher

Cosine rule:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Rearranged to find an angle:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

Use it when you know **two sides and the angle between them** (SAS) or **all three sides** (SSS).

Worked Example: In triangle ABC, $AB = 10$ cm, $AC = 7$ cm and angle $BAC = 52^\circ$. Work out BC .

$$BC^2 = 7^2 + 10^2 - 2 \times 7 \times 10 \times \cos 52^\circ = 149 - 86.19\dots = 62.80\dots$$

$$BC = \sqrt{62.80\dots} = 7.925\dots \approx \mathbf{7.93 \text{ cm}}$$

Worked Example: A triangle has sides 6 cm, 8 cm and 11 cm. Work out its largest angle.

The largest angle is opposite the longest side (11 cm):

$$\cos \theta = \frac{6^2 + 8^2 - 11^2}{2 \times 6 \times 8} = \frac{-21}{96} = -0.21875$$

$$\theta = \cos^{-1}(-0.21875) = \mathbf{102.6^\circ}$$
 — a negative cosine tells you the angle is obtuse.

Exam Tip: Work out $b^2 + c^2$ and $2bc \cos A$ separately before subtracting. Don't use the cosine rule in a right-angled triangle — Pythagoras or SOHCAHTOA is quicker.

4.5.7 Area of a Triangle: $\frac{1}{2}ab \sin C$ (G23) Higher

For any triangle with two sides a and b and the **included** angle C between them:

$$\text{Area} = \frac{1}{2}ab \sin C$$

G23 asks you to use it to find the area, or to work backwards to a side or an angle.

Worked Example: A triangle has sides 9 cm and 13 cm with an included angle of 64° . Work out its area.

$$\text{Area} = \frac{1}{2} \times 9 \times 13 \times \sin 64^\circ = 58.5 \sin 64^\circ = 52.58 \dots \approx \mathbf{52.6 \text{ cm}^2}$$

Worked Example: A triangle has area 50 cm^2 and two sides 10 cm and 14 cm. Work out the possible sizes of the angle between them.

$$50 = \frac{1}{2} \times 10 \times 14 \times \sin C = 70 \sin C \Rightarrow \sin C = \frac{5}{7}$$

$$C = \sin^{-1}\left(\frac{5}{7}\right) = \mathbf{45.6^\circ} \text{ or } C = 180^\circ - 45.6^\circ = \mathbf{134.4^\circ}$$

Choosing the Right Method

Information given	Want	Use
Right angle + two sides	Third side	Pythagoras

Right angle + one side and one angle (or two sides)	Side or angle	SOHCAHTOA
A side with its opposite angle + one more side or angle	Side or angle	Sine rule Higher
Two sides + included angle	Third side	Cosine rule Higher
Three sides	An angle	Cosine rule Higher
Two sides + included angle	Area	$\frac{1}{2}ab \sin C$ Higher

Exam Tip: Edexcel does not give any of these formulae in the exam. Learn Pythagoras, SOHCAHTOA, the exact values, the sine rule, the cosine rule and $\frac{1}{2}ab \sin C$ by heart.